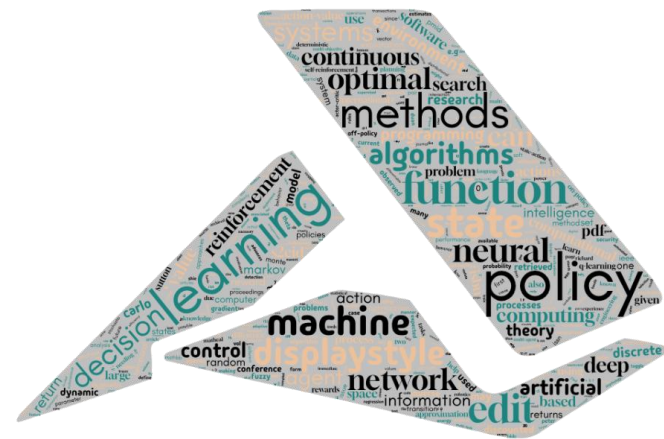




Reinforcement Learning Algorithms and Applications

Chapter Ch 12-3. PPO with Behavior Cloning

Modified Loss Function with Behavior Cloning

- ▶ **The original PPO objective**

- ▶ $L(\theta) = -L^{CLIP} + c_1 L^{VF} - c_2 L^{ENT}$
- ▶ c_1 and c_2 are fixed coefficients throughout training.

- ▶ **The new objective**

- ▶ $L(\theta, t) = w_{PPO}(t)(-L^{CLIP}) + c_1 L^{VF} - c_2 L^{ENT} + w_{BC}(t)L^{BC}$

- ▶ **BC weight $w_{BC}(t)$:**

- ▶ $w_{BC}(t) = \max(w_0 e^{-\alpha t}, w_{floor})$

- ▶ **PPO weight $w_{PPO}(t)$ —starts near zero, rises to a ceiling:**

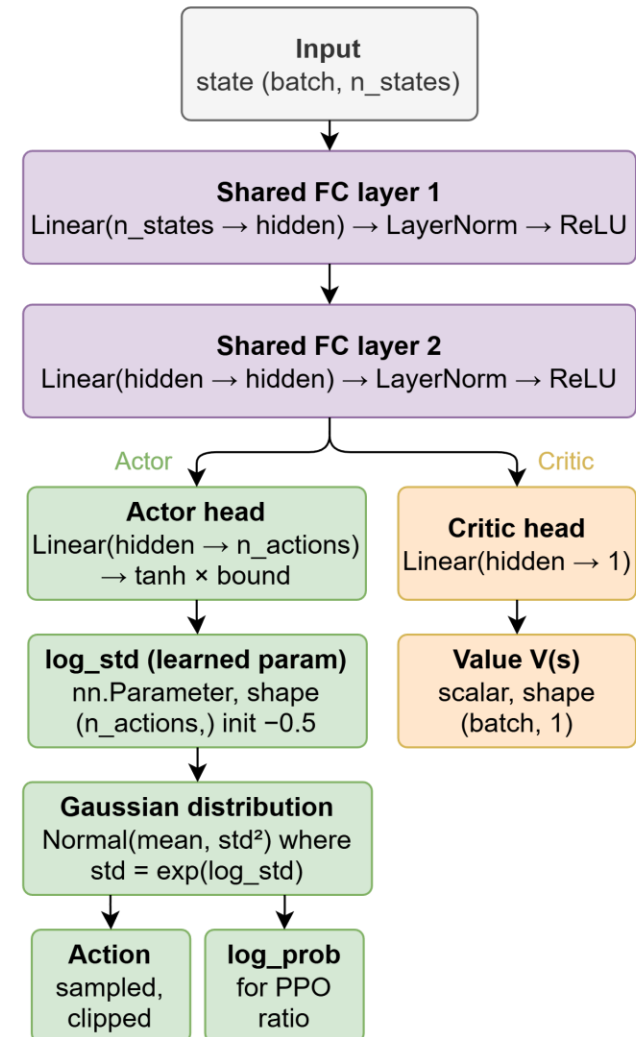
- ▶ $w_{PPO}(t) = w_{max}(1 - e^{-\beta t})$

- ▶ **Behavior cloning loss**

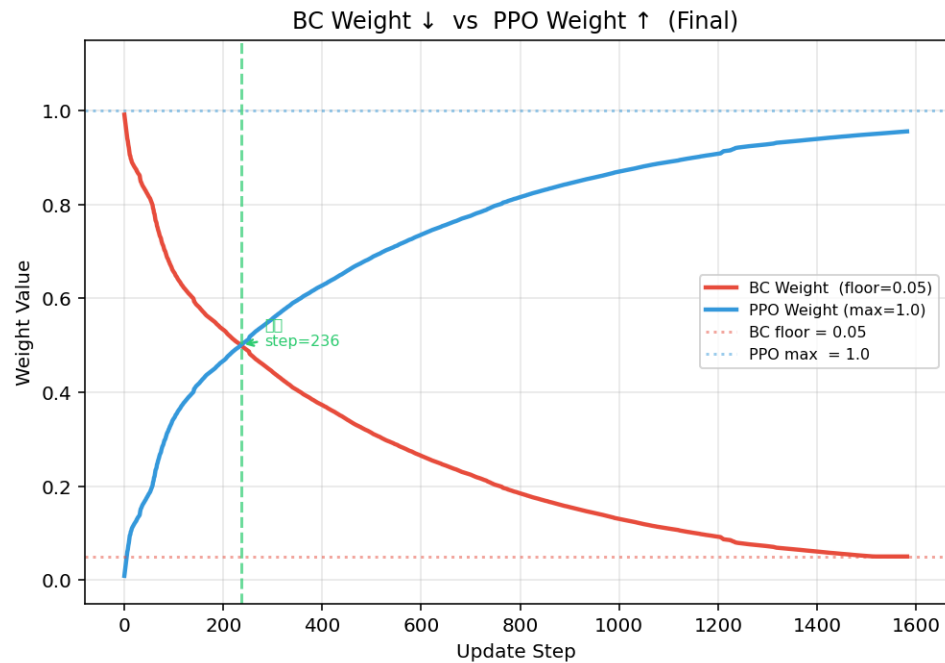
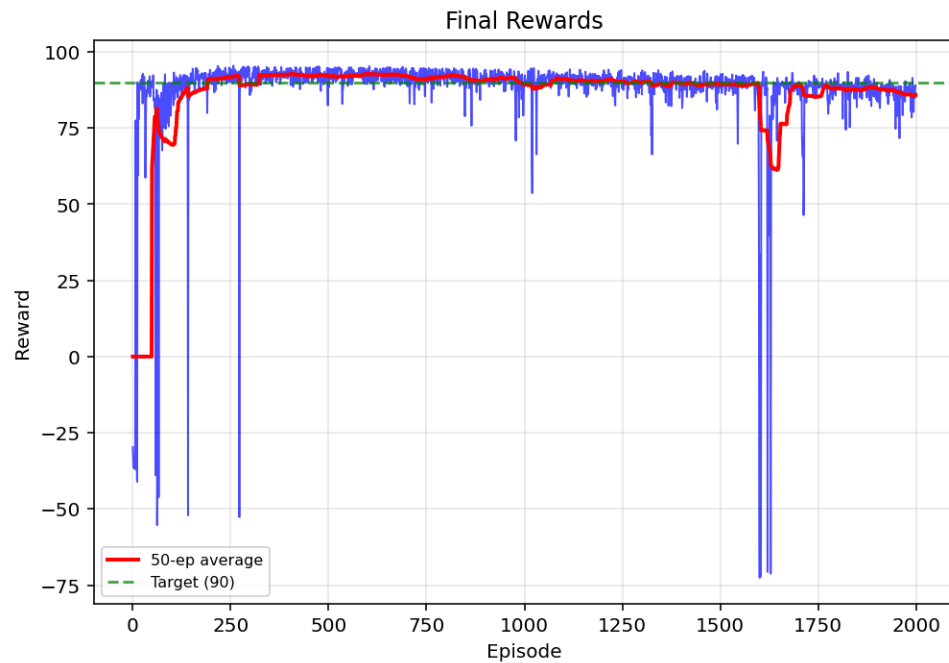
- ▶ $L^{BC} = \text{MSE}(\pi_\theta(s), a_{expert}(s))$

- ▶ **Expert rule**

- ▶ $a_{expert}(s) = \text{sign}(v) a_{max}$



Training Result



Reference

- ▶ M. Lapan, *Deep Reinforcement Learning Hands-On*, 2nd ed. Birmingham, UK: Packt Publishing, 2020.
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- ▶ S. Ravichandiran, *Deep Reinforcement Learning with Python*, 2nd ed. Birmingham, UK: Packt Publishing, 2020.